

Table 2: Evaluation of MCF on Small to medium-sized NETGEN dataset.

Methods	100 × 100		500 × 500		1k × 1k		5k × 5k	
	obj	time	obj	time	obj	time	obj	time
Real(Papadimitriou & Steiglitz)	64.58	421 ms	20.418	1028 s	23.365	2340 s	15.652	18592 s
ZKW(Goldberg et al.)	64.58	504 ms	20.418	706 s	23.365	1639 s	15.652	7491 s
Gurobi(Gurobi Optimization, LLC (2021))	64.58	77.3 s	20.42	4572 s	—	≥ 3 hours	—	≥ 10 hours
EOFT-Sinkhorn	65.05	22.2 s	20.548	82 s	24.696	420 s	16.41	1261 s

Table 3: Evaluation of MCF on very large sparse graph.

Methods	Netgen_8		Netgen_lo_8		Vision	
	obj	time	obj	time	obj	time
pns(k=1,p=1) (Kara & Özturan, 2022)	18.33	335 s	12.88	89 s	30.16	1774 s
pns(k=4,p=4)	18.33	261 s	12.88	71 s	30.16	2579 s
pns(k=16,p=16)	18.33	286 s	12.88	91 s	30.16	2512 s
lenmon-ns (Király & Kovács, 2012)	18.33	186 s	12.88	51 s	30.16	2805 s
EOFT-Sinkhorn	18.83	50.4 s	13.08	59.3 s	32.04	1802 s

5,000 nodes, which we indicate with a "-" symbol. On w/o constraints, there are only flow balance constraints mentioned in Eq. 5. For problems that incorporate both capacity and node constraints, the objective (obj) of the problem slightly increases (e.g., from 10.02 to 10.06) due to the additional constraints, and the computational time increases almost linearly with the problem size. In contrast, our entropy regularization method, specifically the Sinkhorn algorithm (Cuturi, 2013), offers faster solving speeds due to its GPU-friendliness. The results demonstrate that our algorithm significantly outperforms traditional algorithms in terms of efficiency when computing large-scale MCF instances, while maintaining low computational error gap. For further details and parameters regarding the Uniform-MCF, please refer to Table 4 in Appendix D.

Evaluation Results on small to medium-size NETGEN. The experimental details of NETGEN datasets are given in Table 5 and the results are given in Table 2. For smaller problems, Real ((Papadimitriou & Steiglitz)) and ZKW ((Goldberg et al.)) achieve the best solution quality and computational time. However, as the dataset size $|D|$, the total number of nodes $|V|$ and the total number of edges $|E|$ increase, our method demonstrates superior solution speed over former 3 methods. Lemon achieve best solution and speed under all-cases, demonstrated outstanding performance in solving small to medium-scale MCF problems as a SOTA method." Appendix D shows the parameters and details about NETGEN experiments.

Evaluation Results on very large sparse graph. The experimental details for large sparse datasets are presented in Table 6, while the corresponding results are summarized in Table 3. To efficiently solve these three exceptionally large sparse MCF problems, we tailored our algorithm to better leverage the triplet structure inherent in sparse graphs. As the pioneering work applying entropic regularization to the MCF problem and developing a GPU-friendly algorithm, our approach effectively tackles the minimum cost flow problem on extremely large sparse graphs. In certain cases, it even surpasses state-of-the-art algorithms in obtaining high-quality approximate solutions.

4.3 ABLATION STUDY

Then we will analyze the convergence, effect of regularization, and numerical stability in depth.

Convergence of EOFT In Figure 5, we illustrate how the marginal distribution evolves with the number of iterations for an instance with 5000 nodes. The marginal difference is computed as $\mathbf{P}\mathbf{1}_N - \mathbf{P}^\top \mathbf{1}_N$ (denoted as $\mathbf{s}_{\text{soft}}$), while the Ground Truth (GT) marginals are represented by $\mathbf{s}$. Initially, we generate the flow for the source and target separately using a one-dimensional Gaussian function, denoted as $\mathbf{s}_{\text{source}}$ and $\mathbf{s}_{\text{target}}$. To estimate the distribution of the source and target marginals, we employ Gaussian Kernel Density Estimation (KDE) (Parzen, 1962), defined as $K(u) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}u^2}$. The marginal difference at transshipment nodes is obtained by computing the difference between the estimated distributions of the source and target marginals, i.e. $\mathbf{s}_{\text{source}} - \mathbf{s}_{\text{target}}$. The results demonstrate that as the number of iterations increases, EOFT progressively converges to the GT marginals, validating the accuracy and stability of our algorithm. (e.g., the red dotted line (EOFT-200 iterations) in Figure 5 almost overlaps with the solid black line which is GT marginals).